A Project Abstract

on

DEEPFAKE DETECTION FOR IDENTIFYING MACHINE-GENERATED TWEETS USING NLP AND DL

Submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

by

MADHURI E	**214G1A3250**
NADEEM A	**214G1A3257**
CHARAN KUMAR REDDY D	**214G1A3212**
HARSHAVARDHAN REDDY Y	**224G5A3206**

Under the Guidance of

Mrs. G. Shabana, M. Tech.

Department of Computer Science & Engineering (Data Science)

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

(Affiliated to JNTUA, accredited by NAAC with 'A' Grade, Approved by AICTE & Accredited by NBA (EEE, ECE & CSE))

Rotarypuram Village, BK Samudram Mandal, Ananthapuramu - 515701.

2024 – 2025

ABSTRACT

Recent advancements in natural language generation have provided powerful tools capable of manipulating public opinion on social media platforms. The proliferation of text-generative models has enabled adversaries to enhance social bots, generating realistic deepfake posts that influence public discourse. This research addresses the pressing issue of identifying machine-generated text, focusing on social networks like Twitter. The primary objective of this study is to develop reliable and accurate methods for detecting deepfake social media messages.

To achieve this goal, we propose a deep learning-based approach employing a Convolutional Neural Network (CNN) architecture combined with FastText word embeddings to classify tweets as either human-generated or bot-generated. The research utilized the publicly available Tweepfake dataset and compared the performance of the proposed method against several baseline machine learning models. These baselines incorporated features such as Term Frequency (TF), Term Frequency-Inverse Document Frequency (TF-IDF), FastText, and FastText subword embeddings.

The results of the study demonstrated the superior performance of the proposed CNN model with FastText embeddings, highlighting its capability to accurately detect deepfake tweets. These findings underscore the importance of leveraging deep learning and advanced word embeddings for tackling the growing challenge of machine-generated content. This research contributes to the development of robust tools for safeguarding public discourse on social media, with significant implications for combating misinformation and ensuring the integrity of online interactions.

Keywords: *Deepfake detection, Convolutional Neural Network (CNN), Natural Language Processing (NLP), Machine-generated tweets, FastText embeddings.*

Date:

Guide Sign:

Name: Mrs. G. Shabana, M. Tech.